

India EV Charging Network: Operations, Revenue and Performance Analysis (2024–2026)

Project Overview and Objective:

This project analyzes EV charging station operations, energy consumption, revenue generation, connector performance, utilization patterns, and station status across India. Excel was used for data preparation and Power BI was used for dashboard development and visualization.

2. Data Sources :

- Source: EV Charge Station MiniProject.xlsx and Power BI Dashboard. The dataset contains station information including Station ID, City, Operator, Status, Connector Type, Installation Date, Energy Dispensed, and Revenue. The cleaned dataset was imported into Power BI for dashboard creation and business intelligence reporting.

3. Problem Statement (Sample)

Source: EV Charge Station MiniProject.xlsx and Power BI Dashboard. The dataset contains station information including Station ID, City, Operator, Status, Connector Type, Installation Date, Energy Dispensed, and Revenue. The cleaned dataset was imported into Power BI for dashboard creation and business intelligence reporting.

4. Attribute (Column /Features) Details: (Sample)

Attribute Name	Data Type	Description
Station ID	Categorical	Unique Id Number
City	Categorical	Name of the city
Operator	Categorical	Types of Operators
Status	Categorical	Count of Status
Installation Date	Date	Date when the station is installed
Connector Type	Categorical	Types of Connectors
Speed Category	Categorical	Count of speed categorical
Energy Dispensed (kWh)	Numerical	Total Energy Station
Revenue	Numerical	Total Revenue by Station
Year	Date	Years of Installation
Utilization Category	Categorical	Utilization by Connectors
Revenue per kWh	Numerical	Revenue by Energy
Operator Sector	Categorical	Operators By sectors

Connector Region	Categorical	Connectors By Region
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5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
 - **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.
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6. Data Pre-Processing (Excel / Power Query)

Tasks Performed: • Removed duplicate records.

- Standardized city names and operator names.
 - Corrected inconsistent status values.
 - Handled missing values.
 - Created calculated columns such as Year, Revenue per kWh, Utilization Category, Operator Sector, and Connector Region.
 - Loaded transformed data into Power BI using Power Query.
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7. Data Modelling and DAX (Power BI)

- Removed duplicate records.
 - Standardized city names and operator names.
 - Corrected inconsistent status values.
 - Handled missing values.
 - Created calculated columns such as Year, Revenue per kWh, Utilization Category, Operator Sector, and Connector Region.
 - Loaded transformed data into Power BI using Power Query.
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8. Analysis and Visualizations (Power BI)

Dashboard Features:

Excel Analysis:

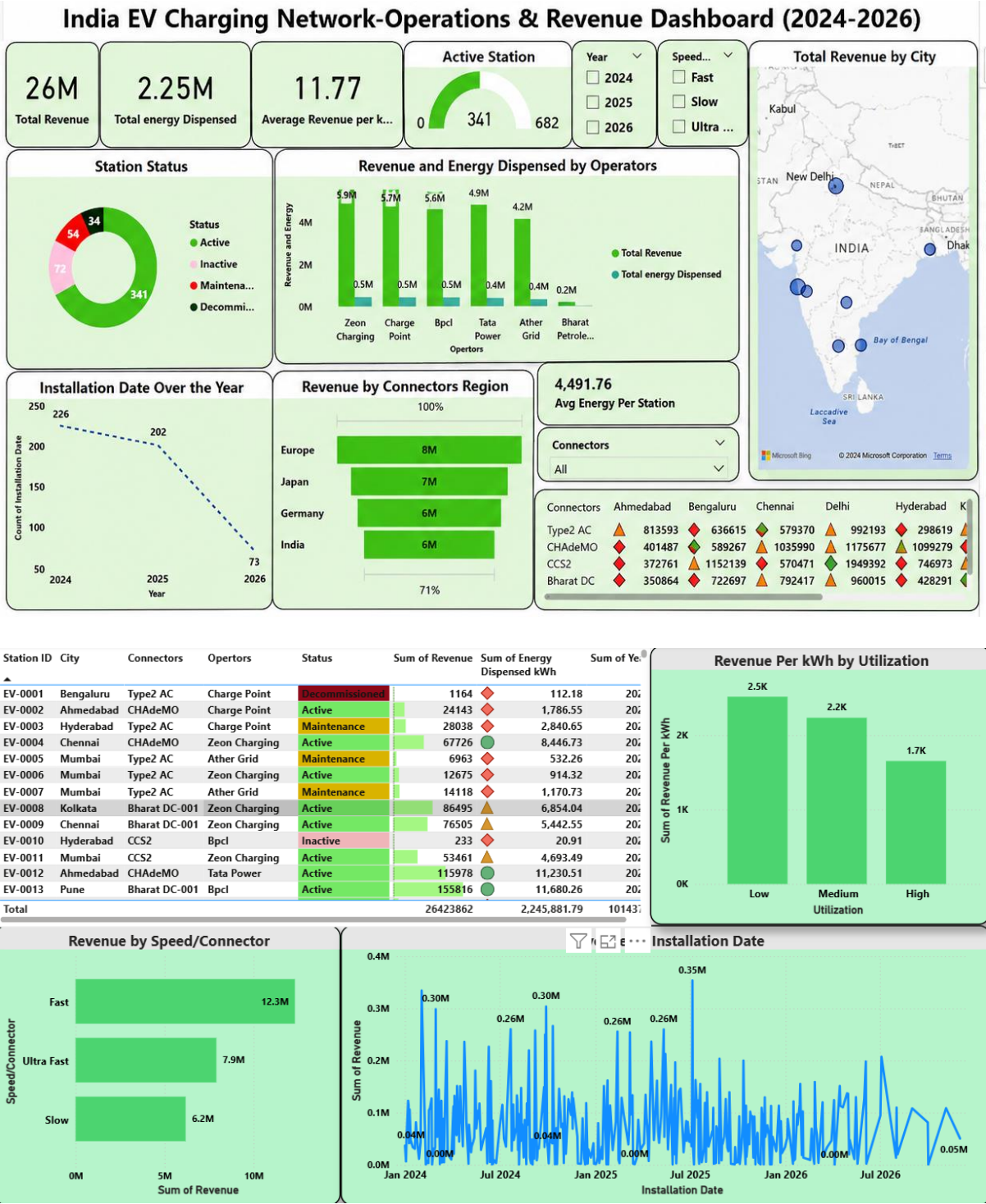
- Data cleaning and validation.
- Revenue and energy calculations.
- Pivot table summaries.

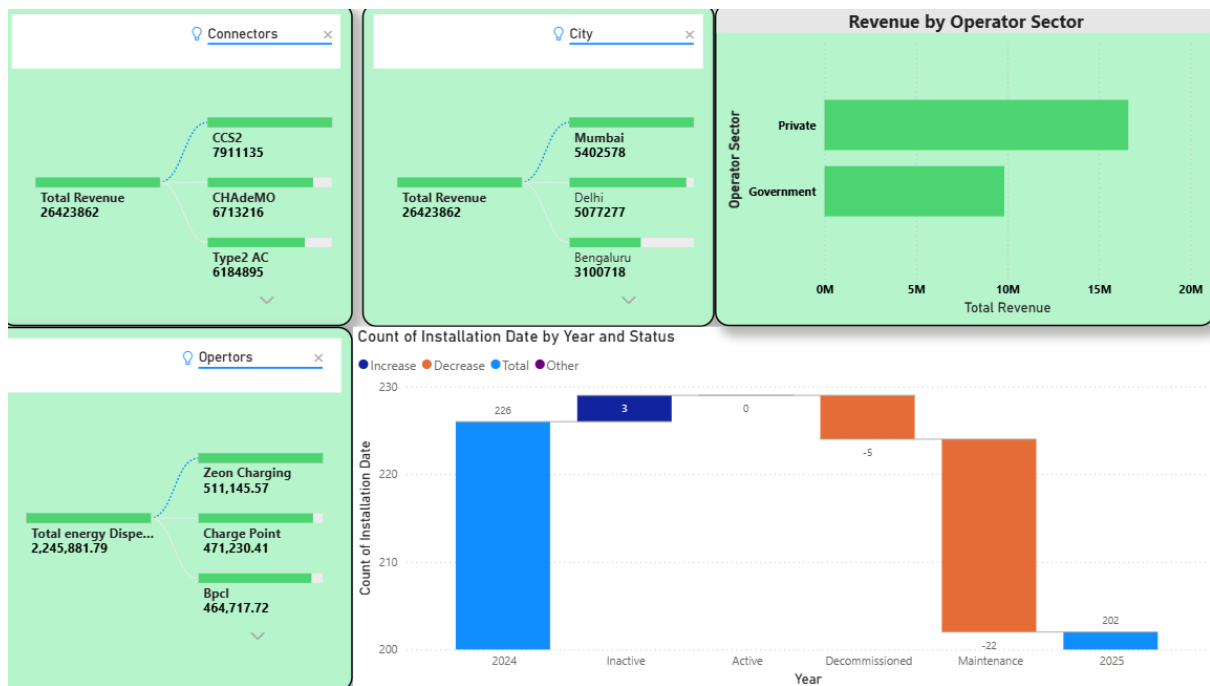
Power BI Dashboard Analysis:

- Total Revenue: 26.42 Million.
- Total Energy Dispensed: 2.25 Million kWh.
- Average Revenue per kWh: 11.77.
- Active Station KPI monitoring.
- Revenue by Operator analysis.
- Revenue by City analysis.
- Revenue by Connector Region analysis.
- Revenue by Speed/Connector analysis.
- Station Status distribution.
- Installation trend analysis (2024–2026).

- Revenue per kWh by Utilization category.
- Station-level performance analysis.

SCREENSHOT OF DASHBOARD /REPORT





9. Insights & Conclusions

- Total revenue exceeded 26 million.
- More than 2.24 million kWh of energy was dispensed.
- Active stations form the majority of the charging network.
- Zeon Charging, Charge Point, and BPCL are leading operators.
- Fast charging connectors generate the highest revenue.
- Mumbai, Delhi, Bengaluru, Chennai, and Hyderabad contribute significantly to revenue.
- Private operators generate more revenue than government operators.
- Installation activity was highest during 2024 and 2025.
- CCS2, CHAdeMO, and Type2 AC are major connector contributors.

10. Conclusions (Sample)

The project successfully integrates Excel-based data preparation with Power BI analytics to evaluate India's EV charging ecosystem. The analysis provides valuable insights into revenue trends, station performance, operator contribution, energy utilization, and connector effectiveness. These findings support strategic planning and future expansion of EV charging infrastructure.